

Written Exam Economics winter 2025-26

Programming for Economists

February 20 to February 22

This exam question consists of 7 pages in total

Answers only in English.

You should hand-in a single zip-file named with your groupname only. The zip-file should contain:

1. A general README.md for your portfolio
2. Your data analysis project (in the folder 01_dataproject)
3. Your model analysis project (in the folder 02_modelproject)
4. Your exam project (in the folder 03_examproject)

Use of GAI tools is permitted. You must explain how you have used the tools. You are required to complete the GAI declaration and upload it as an attachment when submitting your assignment. If you do not submit the GAI declaration, your exam submission will be rejected, and you will have used an exam attempt.

When text is solely or mainly generated by a GAI tool, the tool used must be quoted as a source.

Be careful not to cheat at exams!

Exam cheating is for example if you:

- Copy other people's texts without making use of quotation marks and source referencing, so that it may appear to be your own text
- Use the ideas or thoughts of others without making use of source referencing, so it may appear to be your own idea or your thoughts
- Reuse parts of a written paper that you have previously submitted and for which you have received a pass grade without making use of quotation marks or source references (self-plagiarism)
- Receive help from others in contrary to the rules in the Faculty of Social Science's common part of the curriculum

You can read more about the rules on exam cheating on your Study Site and in the Faculty of Social Science's common part of the curriculum.

Exam cheating is always sanctioned by a written warning and expulsion from the exam in question. In most cases, the student will also be expelled from the University for one semester.

1 Danish Consumption

1.1 Consumption-income ratio

Use an API call to download quarterly data on household disposable income and consumption from table NKS02 at Statistics Denmark. The disposable income variable is B6GD, the consumption variable is P3D, and the sector variable is S1M. Plot the ratio of consumption to disposable income over time. Additionally, seasonally adjust the consumption and income series using `statsmodels.tsa.seasonal.STL`, and plot the seasonally adjusted ratio. Add horizontal lines for the mean in three sub-periods, i) before 2008Q1, ii) from 2010Q1 to 2019Q4, iii) from 2021Q1 to the latest available observation.

1.2 Consumption shares

Use an API call to download consumption data from table NKHC021 at Statistics Denmark in current prices and adjusted for seasonality. Compare total consumption in this table with the consumption measure used in Question 1.1. Compute and plot consumption shares in one or more figures, which are easy to interpret.

1.3 Change in consumption shares

Compute the change in each consumption share between the average in 1990 and the average of available quarters in 2025. Make a bar chart of the increases and decreases in percentage points for all categories. Repeat the exercise computing the change since average in 2019.

2 Labor supply with non-linear wage schedule

A worker chooses labor supply $\ell \in [0, \ell_{\max}]$. Consumption is denoted c .

Preferences are

$$u(c, \ell) = \frac{c^{1-\sigma} - 1}{1-\sigma} - \psi \frac{\ell^{1+\eta}}{1+\eta}, \quad \sigma > 0, \sigma \neq 1, \psi > 0, \eta > 0. \quad (1)$$

Throughout the problem use the following parameter values:

Non-labor income	y_0	0.2
Base wage	w	1.0
Risk aversion	σ	2.0
Disutility scale	ψ	1.0
Curvature of disutility	η	0.5
Maximum hours	$\ell_{\max}$	1.5

You can use the code in `LaborSupplyModel.py` uploaded together with the exam as your starting point.

2.1 Baseline (linear wage)

In the baseline system, the worker receives a constant wage w per hour:

$$c(\ell) = y_0 + w\ell. \quad (2)$$

The first order condition is

$$wc^{-\sigma} = \psi\ell^\eta \quad (3)$$

1. Solve for ℓ^* with the three methods below and report (ℓ^*, c^*, u^*) :

- (a) A grid search
- (b) A local optimizer
- (c) A root finder using the first order condition

2. Compute $u(\ell) \equiv u(c(\ell), \ell)$ on a grid $\ell \in [0, \ell_{\max}]$.

Create a figure with two panels:

- (a) $c(\ell)$ as a function of ℓ .
- (b) $u(\ell)$ as a function of ℓ .

Mark ℓ^* from your favorite solution method as a vertical line.

2.2 Overtime pay

Now the worker receives a higher marginal wage for overtime hours. Let $\bar{\ell}$ denote the overtime threshold. Earnings are

$$\text{earn}(\ell) = \begin{cases} w\ell, & \ell \leq \bar{\ell}, \\ w\bar{\ell} + w(1 + \delta)(\ell - \bar{\ell}), & \ell > \bar{\ell}, \end{cases} \quad (4)$$

and consumption is $c(\ell) = y_0 + \text{earn}(\ell)$. Use $\bar{\ell} = 0.9$ and $\delta = 1.0$.

1. Plot $c(\ell)$ and $u(\ell)$ for $\ell \in [0, \ell_{\max}]$.
2. Use your plot (and/or a grid search) to verify that $u(\ell)$ has *two local interior maxima* for this parameterization.
3. Run a *local* optimizer twice, once with initial guess $\ell_0 = 0.2$ and once with $\ell_0 = 1.2$. Does the initial guess matter? Why?
4. Implement a “robust” solver that always returns the *global* maximizer.
Hint: Solve the problem separately on $[0, \bar{\ell}]$ and $[\bar{\ell}, \ell_{\max}]$, and evaluate the utility value in each region and pick the highest.

3 Labor market simulation

In this assignment you will build and simulate a simple labor-market model with unemployment, human capital dynamics, and two worker types: low-skilled ($s = L$) and high-skilled ($s = H$). The goal is to simulate forward until the cross-sectional distribution does not change anymore, and thus is (approximately) stationary. You will then use the simulation to compare outcomes for low- and high-skilled workers. You can use the code in `LaborMarketModel.py` uploaded together with the exam as your starting point.

State variables For each individual i at time t (think of t as months), the state is $(e_{it}, d_{it}, k_{it}, s_i)$, where:

- $e_{it} \in \{0, 1\}$ is employment status (1 employed, 0 unemployed),
- $d_{it} \in \{0, 1, 2, \dots\}$ is unemployment duration (months), $d_{it} = 0$ if employed,
- k_{it} is *log human capital*,
- $s_i \in \{L, H\}$ is the worker type.

Human capital Log human capital follows an AR(1) process around a type-specific mean, μ_{s_i} , with depreciation, δ_u , during unemployment:

$$k_{i,t+1} = \mu_{s_i} + \rho(k_{it} - \mu_{s_i}) - \delta_u \cdot \mathbf{1}\{e_{it} = 0\} + \sigma_h \eta_{i,t+1}, \quad \eta_{i,t+1} \sim \mathcal{N}(0, 1). \quad (5)$$

Income While employed, income has an idiosyncratic wage shock:

$$y_{it} = \begin{cases} \exp(a_{s_i} + k_{it} + \sigma_y \xi_{it}), & \text{if } e_{it} = 1, \xi_{it} \sim \mathcal{N}(0, 1), \\ b \cdot \exp(a_{s_i} + k_{it}), & \text{if } e_{it} = 0. \end{cases} \quad (6)$$

Here $b \in (0, 1)$ is a replacement rate.

Labor market transitions Employment evolves as follows:

- If employed ($e_{it} = 1$): with probability `sep_rate` the worker separates and becomes unemployed next period, $e_{i,t+1} = 0$, with $d_{i,t+1} = 1$.
- If unemployed ($e_{it} = 0$) with duration $d_{it} \geq 1$: with probability $f_{s_i}(d_{it})$ the worker finds a job and becomes employed next period ($e_{i,t+1} = 1$ and $d_{i,t+1} = 0$). Otherwise they remain unemployed and duration increases by one: $d_{i,t+1} = d_{it} + 1$.

Job-finding rate with duration dependence The job-finding probability declines with unemployment duration,

$$f_s(d) = \max \left\{ \min \left\{ 1, f_{0,s} \exp \left(-\lambda_s(d-1) \right) \right\}, 0.05 \right\}, \quad d \geq 1, \quad (7)$$

but never falls below 0.05.

Initialization (given in the code) To start the simulation at $t = 0$, use a simple random initialization:

- Draw types: $s_i = H$ with probability 0.40 (otherwise $s_i = L$).
- Draw initial employment: Employed with probability 0.92.
- If employed, set initial duration: $d_{i0} = 0$.
- If unemployed draw d_{i0} uniformly from $\{1, 2, 3, 4, 5, 6\}$.
- Draw initial human capital from a stationary distribution of the AR(1) around μ_{s_i} (ignoring unemployment depreciation at initialization):

$$k_{i0} = \mu_{s_i} + \frac{\sigma_h}{\sqrt{1-\rho^2}} z_i, \quad z_i \sim \mathcal{N}(0, 1).$$

This initialization is already implemented for you in the provided `setup_states()` function.

Calibration Use the following baseline parameters:

	Low-skilled (L)	High-skilled (H)
Human capital mean μ_s	0.00	0.50
Wage intercept a_s	2.40	2.80
Baseline finding rate $f_{0,s}$	0.25	0.45
Duration slope λ_s	0.08	0.03

Common parameters:

$$\rho = 0.97, \quad \sigma_h = 0.06, \quad \delta_u = 0.03, \quad \sigma_y = 0.20, \quad b = 0.60, \quad \text{sep_rate} = 0.02.$$

Simulate a population of $N = 50,000$ individuals, with a high-skilled share $\Pr(s_i = H) = 0.40$. Use the fixed random seed 123 to make your results reproducible.

Simulation Simulate for $T = 3,000$ periods. Your simulator must *store only the last 200 periods* (think of this as a small “panel” of stationary data). Use this stored panel to answer the questions below.

3.1 Document stationarity

Run the model with the baseline parameters. Let the simulator return the last 200 periods and denote these periods by $t = -200, \dots, -1$ (where $t = -1$ is the final simulated period).

1. Compute the selected moments (unemployment rate, mean and standard deviation of $\log y$) using the first half ($t = -200, \dots, -101$) and the second half ($t = -100, \dots, -1$). Show that the moments are (almost) identical.
2. Compare the distribution of $\log y$ and the distribution of unemployment duration among the unemployed for the two halves ($-200 : -100$ vs. $-100 :$). Argue that the distributions are the same.

3.2 Compare low- vs high-skilled

Using the stored stationary panel (you may pool all 200 periods), compare outcomes for low- and high-skilled workers. Report and discuss at least: the unemployment rate by type, the mean and standard deviation of $\log y$ by type, and a summary of the duration distribution by type (e.g. mean duration among the unemployed and the share unemployed for more than 12 months).

3.3 Reasons for inequality

The type-specific parameters are $(\mu_s, a_s, f_{0,s}, \lambda_s)$. Analyze which of these are most important for the differences between low- and high-skilled workers. A simple approach is to run counterfactual simulations where you change *one* type-specific parameter at a time (e.g. set the high-skilled value equal to the low-skilled value) keeping the other parameters fixed, and measure how the gaps in outcomes change. Summarize your findings in a small table.